

PowerNext-AI Screening Submission: Methodology

1. Problem Statement and Objective

The objective of this solution is to develop an automated machine-learning pipeline for analyzing electrical test data and predicting two key outcomes for unseen test cases:

- The Reference Parameter associated with each test.
- The Validity Label indicating whether the test is Valid or Invalid.

The provided dataset consists of a labeled training dataset and an unlabeled test dataset. The Training_Data sheet contains 1,000 labeled observations, while the Test_Data sheet contains 350 observations for which predictions are required.

The Reference_Parameter is treated as a continuous regression target, while Validity_Label is treated as a binary classification target. Test_ID is used only to identify test records in the final submission and is excluded from model training.

2. Data Understanding and Preprocessing

The input variables used by the machine-learning models are:

- Applied_Voltage_kV
- Load_Current_A
- Ambient_Temperature_C
- Test_Duration_min
- Sensor_S1
- Sensor_S2
- Sensor_S3
- Sensor_S4

These features represent the electrical operating conditions, environmental conditions, test duration, and sensor measurements associated with each test.

The training dataset was checked for missing values and duplicate records. Missing numerical values were handled automatically using median imputation within the machine-learning pipeline. Missingness indicators were also included so that the models could retain information about whether a measurement was originally missing.

Additional derived features were generated from the sensor and operating variables. These included:

- Mean of the sensor measurements
- Standard deviation of the sensor measurements
- Sensor measurement range
- Voltage-current power proxy
- Temperature-duration interaction proxy

Exact duplicate records were checked and none were found. No individual rows were manually modified or removed. This ensures that the complete process remains reproducible and can be executed automatically for the entire dataset.

3. Reference Parameter Prediction

The Reference Parameter prediction task was formulated as a supervised regression problem.

An ExtraTreesRegressor was selected as the regression model. The model uses an ensemble of randomized decision trees and is suitable for this dataset because it can capture nonlinear relationships and interactions between electrical operating conditions and sensor measurements without requiring a predefined physical equation.

The selected model uses:

- 400 trees
- Minimum leaf size of 2
- 90% feature subsampling

Model evaluation was performed using 5-fold shuffled cross-validation with a fixed random seed of 42.

The resulting cross-validation performance was:

Metric	Result
MAE	0.9340
RMSE	2.3522
R ²	0.9516

The R² value of approximately 0.952 indicates that the model explains a substantial proportion of the variation in the Reference Parameter within the validation data.

4. Validity Classification and Abnormal Data Detection

The Validity prediction task was formulated as a binary classification problem using the Validity_Label available in the training dataset.

An ExtraTreesClassifier was used to learn the relationship between the observed operating conditions, sensor measurements, and the corresponding validity label.

The classifier uses:

- 400 trees
- Minimum leaf size of 3
- Balanced class weighting

Five-fold stratified cross-validation was used so that the class distribution was appropriately maintained across validation folds.

Instead of using the default classification probability threshold of 0.50, a threshold of 0.35 was selected using out-of-fold validation probabilities. This threshold was selected without using the test dataset, preventing test-data leakage.

The validation results were:

Metric	Result
Accuracy	0.9460
Invalid Precision	0.7985
Invalid Recall	0.7985
Invalid F1-score	0.7985

A residual-based approach using the regression model was also investigated for identifying abnormal observations. However, its signal was weaker than the supervised classification approach. Therefore, the supervised classifier was retained as the primary method for predicting Validity_Label.

5. Validation Strategy and Assumptions

The complete pipeline uses a fixed random seed to ensure reproducibility. Regression uses shuffled 5-fold cross-validation, while classification uses stratified 5-fold cross-validation.

The following assumptions were made:

- The observations are treated as independent because no repeated-group identifier was provided.
- The feature definitions and units in the test dataset are consistent with those in the training dataset.
- Missing numerical sensor values can be reasonably represented using median imputation.
- Extreme observations are retained rather than manually deleted, allowing the ensemble models to handle nonlinear and unusual operating conditions.
- The labels in the training dataset are assumed to be reliable.
- Validation metrics represent estimates on held-out training data and should not be interpreted as guaranteed performance on the hidden evaluation dataset.

The complete workflow is automated, and no individual test record is manually modified.

6. Automated Prediction and Submission Generation

After training the models on the available labeled training data, the final models are applied to all 350 test records.

For each test case, the pipeline generates:

- Test_ID
- Predicted_Reference_Parameter
- Validity_Label

The final predictions are automatically written to the required submission CSV file. The pipeline also generates an automated summary.json containing dataset information, validation metrics, prediction statistics, model details, and environment information.

This ensures that the solution can be executed end-to-end without manual intervention or individual record modification.

7. Digital Twin Automation

The proposed solution can be extended into a digital-twin-based monitoring system for real-world deployment.

The overall workflow can be represented as:

Physical Equipment → Sensors → Data Acquisition → Validation Gateway → Digital Twin → ML Models → Prediction & Anomaly Detection → Operator Alerts

In a deployed system, equipment sensors would continuously provide timestamped measurements to a data-acquisition layer. A validation gateway would check the incoming data for correct schema, units, valid ranges, missing measurements, and duplicate events.

The validated data would then pass through the same locked preprocessing and feature-engineering pipeline used during model development. The digital twin would maintain an estimate of the current operating state while the machine-learning models would:

- Predict the Reference Parameter.
- Predict the Validity_Label.
- Identify potentially abnormal operating conditions.
- Provide information to support operator decision-making.

Additional monitoring components can track model confidence, data drift, and sensor health. Confirmed outcomes from the physical system can be stored and used for periodic model evaluation, recalibration, champion/challenger testing, and controlled retraining.

This provides a pathway from the current offline screening model toward a continuously monitored and automated digital-twin system.

8. Conclusion

The proposed solution combines automated preprocessing, feature engineering, ensemble regression, and ensemble classification into a reproducible end-to-end pipeline. The regression model provides Reference Parameter predictions, while the classification model provides automated Validity_Label predictions.

The use of cross-validation, a fixed random seed, automated preprocessing, and test-independent threshold selection helps improve the reliability and reproducibility of the solution. The same pipeline can subsequently be integrated with a digital twin to support real-time monitoring, abnormal-data detection, and controlled model improvement.